Scenario

In this lab, I am a security analyst who must monitor traffic on my employer's network. I'll be required to configure Suricata and use it to trigger alerts. I have been given a sample pcap file.

Here is how I have done it:

Task 1. Examine a custom rule in Suricata

To do this, I ran following commands:

- **/home/analyst**
- **/var/log/suricata** directory to locate files
- **ls** to bring the file list
- **cat custom.rules** to display the rule in the **custom.rules** file

```
analyst@97dbf040f96b:~$ ls
custom.rules  sample.pcap
analyst@97dbf040f96b:~$ cat custom.rules
alert http $HOME_NET any -> $EXTERNAL_NET any (msg:"GET on wire"; flow:e
stablished,to_server; content:"GET"; http_method; sid:12345; rev:3;)
analyst@97dbf040f96b:~$ ls -l /var/log/suricata
total 0
analyst@97dbf040f96b:~$ sudo suricata -r sample.pcap -S custome.rules -k
 none
15/2/2026 -- 08:15:05 - <Notice> - This is Suricata version 6.0.1 RELEAS
E running in USER mode
15/2/2026 -- 08:15:06 - <Warning> - [ERRCODE: SC_ERR_NO_RULES(42)] - No
rule files match the pattern custome.rules
15/2/2026 -- 08:15:06 - <Warning> - [ERRCODE: SC_ERR_NO_RULES_LOADED(43)
] - 1 rule files specified, but no rules were loaded!
15/2/2026 -- 08:15:06 - <Notice> - all 2 packet processing threads, 4 ma
nagement threads initialized, engine started.
15/2/2026 -- 08:15:06 - <Notice> - Signal Received.  Stopping engine.
15/2/2026 -- 08:15:06 - <Notice> - Pcap-file module read 1 files, 200 pa
ckets, 54238 bytes
analyst@97dbf040f96b:~$ ls -l /var/log/suricata
total 8
-rw-r--r-- 1 root root    0 Feb 15 08:15 eve.json
-rw-r--r-- 1 root root    0 Feb 15 08:15 fast.log
-rw-r--r-- 1 root root 2768 Feb 15 08:15 stats.log
-rw-r--r-- 1 root root 1652 Feb 15 08:15 suricata.log
analyst@97dbf040f96b:~$
```

Here, the rules consist of three components: an action, a header and rule options. The breakdown is -

- ➢ **alert** bit is an **action**

- **http $HOME_NET any -> $EXTERNAL_NET any** is a **header** (has protocol, source and destination IP addresses, source and destination ports and traffic direction).
- **msg** bit provides the alert text: **Get on wire**.
- The **flow:established, to_server** option determines that the packets from the client to the server should be matched.
- **content:"Get"** option tell Suricata to look for the word **GET** in the content of the **http.method** portion of the packet.
- **sid:12345** is the **signature ID** that defines the rule.
- The **rev:3** bit is revision version 3.

To summarise, this signature triggers an alert whenever Suricata observes the text **GET** as the HTTP method in an HTTP packet from the home network going to the external network.

Task 2. Trigger a custom rule in Suricata

I ran command ***ls –l /var/log/suricata*** to list the files. Here, I can see that there is no file in this directory at his point. Next, I ran ***sudo suricata –r sample.pcap -S custom.rules -k none*** where ***–r sample.pcap*** specifies an input file to mimic network traffic, ***-S custom.rules*** tells Suricata to use rules defined in that file and ***–k none*** instructs Suricata to disable all checksum checks. Next, I ran the command ***ls –l /var/log/suricata*** again to list the files in /var/log/suricata folder.

After that, I used **cat** to display the **fast.log** file. Each line or entry in the **fast.log** file corresponds to an alert generated by Suricata when it processes a packet that meets the conditions of an alert generating rule. Each alert line includes the message that identifies the rule that triggered the alert, as well as the source, destination, and direction of the traffic.

```
analyst@97dbf040f96b:~$ cat /var/log/suricata/fast.log
11/23/2022-12:38:34.624866  [**] [1:12345:3] GET on wire [**] [Classific
ation: (null)] [Priority: 3] {TCP} 172.21.224.2:49652 -> 142.250.1.139:8
0
11/23/2022-12:38:58.958203  [**] [1:12345:3] GET on wire [**] [Classific
ation: (null)] [Priority: 3] {TCP} 172.21.224.2:58494 -> 142.250.1.102:8
0
analyst@97dbf040f96b:~$ 
```

Task 3. Examine eve.json output

I have used ***cat /var/log/suricata/eve.json*** command to display the entries in that file.

```
analyst@97dbf040f96b:~$ cat /var/log/suricata/eve.json
{"timestamp":"2022-11-23T12:38:34.624866+0000","flow_id":193185821915285
,"pcap_cnt":70,"event_type":"alert","src_ip":"172.21.224.2","src_port":4
9652,"dest_ip":"142.250.1.139","dest_port":80,"proto":"TCP","tx_id":0,"a
lert":{"action":"allowed","gid":1,"signature_id":12345,"rev":3,"signatur
e":"GET on wire","category":"","severity":3},"http":{"hostname":"opensou
rce.google.com","url":"/","http_user_agent":"curl/7.74.0","http_content_
type":"text/html","http_method":"GET","protocol":"HTTP/1.1","status":301
,"redirect":"https://opensource.google/","length":223},"app_proto":"http
","flow":{"pkts_toserver":4,"pkts_toclient":3,"bytes_toserver":357,"byte
s_toclient":788,"start":"2022-11-23T12:38:34.620693+0000"}}
{"timestamp":"2022-11-23T12:38:58.958203+0000","flow_id":169963486514917
2,"pcap_cnt":151,"event_type":"alert","src_ip":"172.21.224.2","src_port"
:58494,"dest_ip":"142.250.1.102","dest_port":80,"proto":"TCP","tx_id":0,
"alert":{"action":"allowed","gid":1,"signature_id":12345,"rev":3,"signat
ure":"GET on wire","category":"","severity":3},"http":{"hostname":"opens
ource.google.com","url":"/","http_user_agent":"curl/7.74.0","http_conten
t_type":"text/html","http_method":"GET","protocol":"HTTP/1.1","status":3
01,"redirect":"https://opensource.google/","length":223},"app_proto":"ht
tp","flow":{"pkts_toserver":4,"pkts_toclient":3,"bytes_toserver":357,"by
tes_toclient":797,"start":"2022-11-23T12:38:58.955636+0000"}}
analyst@97dbf040f96b:~$
```

The output here is raw content of that eve.json file. So, I ran command ***jq . /var/log/suricata/eve.json | less*** to display entries in an improved format.

```
analyst@97dbf040f96b:~$ jq . /var/log/suricata/eve.json | less
{
  "timestamp": "2022-11-23T12:38:34.624866+0000",
  "flow_id": 193185821915285,
  "pcap_cnt": 70,
  "event_type": "alert",
  "src_ip": "172.21.224.2",
  "src_port": 49652,
  "dest_ip": "142.250.1.139",
  "dest_port": 80,
  "proto": "TCP",
  "tx_id": 0,
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 12345,
    "rev": 3,
    "signature": "GET on wire",
    "category": "",
    "severity": 3
  },
  "http": {
    "hostname": "opensource.google.com",
    "url": "/",
    "http_user_agent": "curl/7.74.0",
    "http_content_type": "text/html",
    "http_method": "GET",
    "protocol": "HTTP/1.1",
```

Next, I used ***jq –c "[.timestamp,.flow_id,.alert.signature,.proto,.dest_ip]" /var/log/suricata/eve.json*** command to extract specific event data (specified in the square brackets) from that eve.json file:

```
analyst@350c67bd479c:~$ jq -c "
> [.timestamp,.flow_id,.alert.signature,.proto,.dest_ip]"
/var/log/suricata/eve.json
parse error: Invalid numeric literal at line 2, column 0
analyst@350c67bd479c:~$ jq -c "[.timestamp,.flow_id,.alert.signature,.pr
oto,.dest_ip]" /var/log/suricata/eve.json
["2022-11-23T12:38:34.624866+0000",2193111573493909,"GET on wire","TCP",
"142.250.1.139"]
["2022-11-23T12:38:58.958203+0000",542577118057716,"GET on wire","TCP","
142.250.1.102"]
analyst@350c67bd479c:~$ jq "select(.flow_id==X)" /var/log/suricata/eve.j
son
jq: error: X/0 is not defined at <top-level>, line 1:
select(.flow_id==X)
jq: 1 compile error
analyst@350c67bd479c:~$
```

Finally, as part of the task, I have used **jq** command to display all event logs related to a specific **flow_id** from the **eve.json** file. The **flow_id** is very useful as it correlates network traffic that belong to the same network flows. By this, I finished my lab as I configured Suricata to trigger alerts on network traffic.

```
analyst@bd31efce953c:~$ jq "select(.flow_id==1169891449796757)" /var/log
/suricata/eve.json
{
  "timestamp": "2022-11-23T12:38:34.624866+0000",
  "flow_id": 1169891449796757,
  "pcap_cnt": 70,
  "event_type": "alert",
  "src_ip": "172.21.224.2",
  "src_port": 49652,
  "dest_ip": "142.250.1.139",
  "dest_port": 80,
  "proto": "TCP",
  "tx_id": 0,
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 12345,
    "rev": 3,
    "signature": "GET on wire",
    "category": "",
    "severity": 3
  },
  "http": {
    "hostname": "opensource.google.com",
    "url": "/",
    "http_user_agent": "curl/7.74.0",
    "http_content_type": "text/html",
```